1 Experimental Results

Table 1: Effect of threshold values

OUR Grouping OPERATORS	$\tau = 0.6$	$\tau = 0.7$	$\tau = 0.8$	$\tau = 0.9$
Grouping 1	0.9903	0.9923	0.9937	0.9960
Grouping 2	0.9783	0.9853	0.9897	0.9930
Grouping 3	0.9923	0.9933	0.9950	0.9970
Grouping 4	0.9840	0.9890	0.9927	0.9940

Table 2: Accuracy, F1 and Sensitivity and other parameters

Method	**Accuracy**	**F1**	**Sensitivity**
Best Grouping (Ours)	0.9970	0.9970	0.9970
KNN (Single Classifier)	0.9957	0.9957	0.9957
SVM (Single Classifier)	0.9777	0.9777	0.9777
RF (Single Classifier)	0.9840	0.9840	0.9840
LR (Single Classifier)	0.9777	0.9777	0.9777
XGBoost (Single Classifier)	0.9927	0.9927	0.9927

Table 3: Comparison with state-of-the-art method

Method	**Accuracy**	**F1**	**Sensitivity**
Masud et al. (2021)	0.9633	0.9630	0.9633
Mangal et al. (2020)	0.9789	0.9785	0.9789
Nishio et al. (2021)	0.9500	0.9490	0.9500
Hatuwal & Thapa (2020)	0.9720	0.9715	0.9720
Talukder et al. (2022)	0.9810	0.9808	0.9810
Hage Chehade et al. (2022)	0.9625	0.9620	0.9625
Proposed Method (Ours)	0.9970	0.9970	0.9970

Table 4: Effect of number of generations and the parameters (you can add other parameters)

No. of Generations	**Accuracy**	**Sensitivity**
50	0.9957	0.9957
100	0.9967	0.9967
150	0.9967	0.9967

Table 5: Effect of crossover value

Crossover Value	**Accuracy**	**Sensitivity**
0.4	0.9957	0.9957
0.5	0.9960	0.9960
0.6	0.9957	0.9957

Table 6: Other parameters of GA

Parameter	**Value**
Accuracy	0.9970
Sn	0.9970

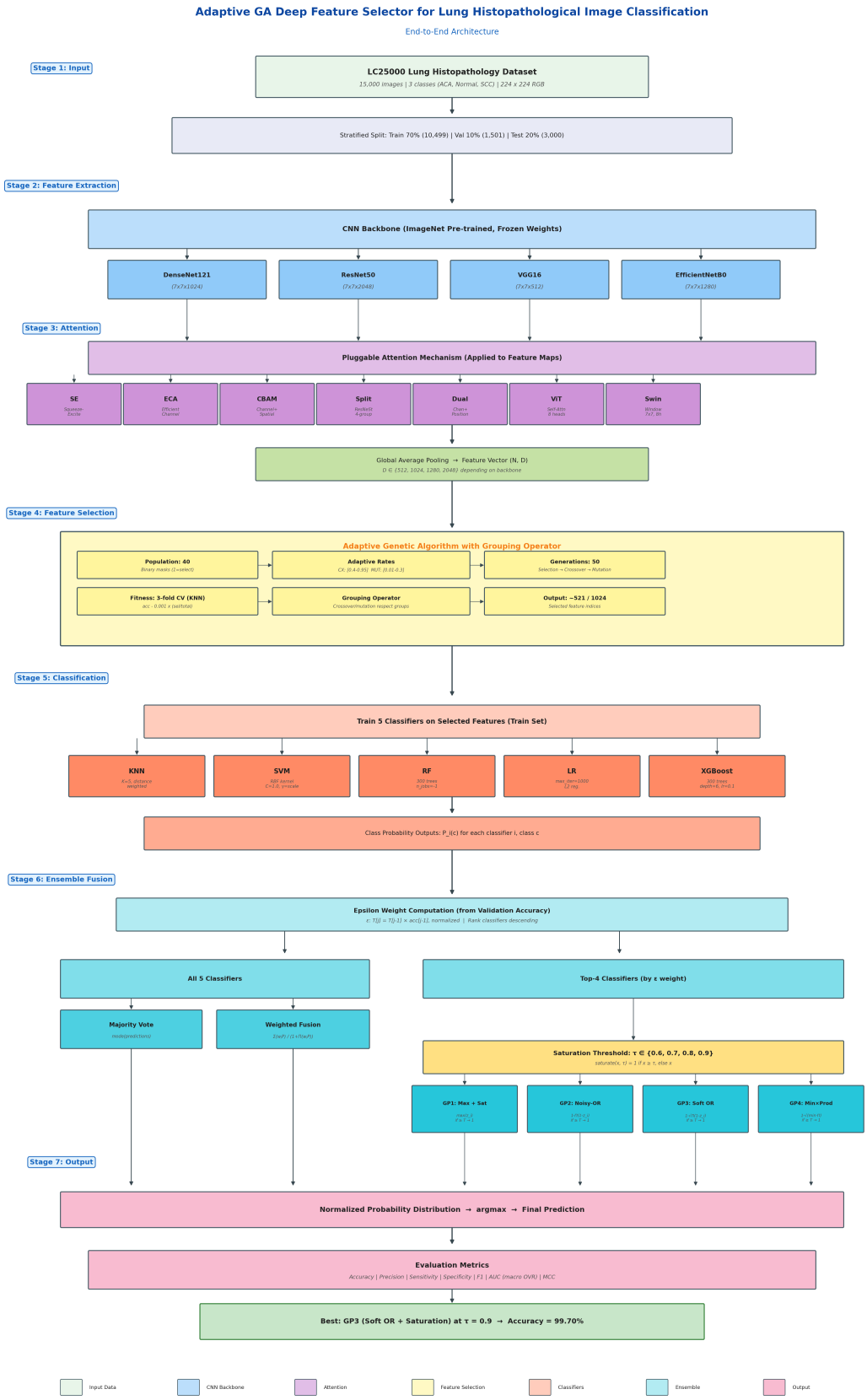

Figure 1: Proposed Architecture: Adaptive GA Deep Feature Selector for Lung Histopathological Image Classification